

Status

Boba — Feature Status (all components)

Revision: 7 **Last modified:** 2026-06-16T23:30:00Z **Scope:** Every system component, service, infrastructure piece, and client app of the Boba project — one row per REAL unit (endpoint / handler / client method / component control / plugin / subcommand / script), grouped by component. **Authority:** assembled by READ-ONLY repo inventory (codegraph + grep + source reading) on 2026-06-15, expanded to per-unit granularity (§11.4.118 discovery-pressure) on 2026-06-16. Cross-references AGENTS.md, CLAUDE.md, docs/REMAINING_WORK_PLAN.md.

Rev 7 (2026-06-16) — live-PASS update for the search/auth fix batch (HEAD 7e9cab5): the multi-word URL-encoding fix (da7d709, ~17 nova3 plugins), the new RUTRACKER_COOKIES injection (2fc29fc), NNMCLUB_COOKIES auth, /auth/status cookie reflection (9c2f8dc), the new GET /api/v1/healthz JSON endpoint (137d7ff), and §11.4.85 stress+chaos coverage for the multi-word fix (7e9cab5) were all PROVEN end-to-end on the live nezha stack — full-fleet the matrix = **2600 results / 23 contributing trackers / zero plugin_bad_query_encoding**, all four private trackers authenticate (rutracker 50, nnmclub 50, kinozal 50, iptorrents 49). Evidence: docs/qa/search-fix-verify-20260616/. kickass.py reclassified **Won't-fix structurally-impossible (§11.4.112)** per docs/research/kickass_403_20260616/. NOTE (§11.4.6): no Go files were touched this session — the Go rows are unchanged (the pre-existing magnet_stress_chaos_test.go guard stands; no NEW Go query-encoding guard was added this session, so none is claimed here).

Captured-evidence-driven (§11.4.5 / §11.4.45 / §11.4.86). Every feature row cites a real source file (file:line where load-bearing), endpoint, command, or control. **No invented features** (§11.4.6). As of Rev 5 the **Video** column is DEFINITIVE for every row — there is no bare PENDING. Each cell is exactly one of: VIDEO-CONFIRMED – <file> (the feature is

shown on-screen in a committed recording), PENDING (UI – film next) (a user-visible control/dialog not yet *individually* filmed), or N/A (no UI – test-covered + exercised by <journey>) (a non-user-visible unit — REST endpoint / handler / client method / parser / scanner / service / crypto / repo / script — which has no screen of its own and is confirmed by its cited tests plus the UI/CLI journey that drives it). **Honest classification (§11.4.6/§11.4.143): a row is VIDEO-CONFIRMED only when a real recording actually shows it.**

Per-row Video tally (288 feature rows): **28 VIDEO-CONFIRMED, 14 PENDING (UI – film next), 246 N/A (no UI – test-covered).**

Column legend: - **Impl** = implemented / partial / stub / missing - **Wired** = reachable on a running default-profile stack (yes / no / opt-in / host) - **Tests** = test types present + cite file (unit / integration / e2e / security / stress-chaos / property / contract); none if no covering test found - **Validation** = honest validation posture (tested-green-in-suite / unit-only / live-evidence-captured / not-validated / operator-blocked) - **Video** = DEFINITIVE video-recording status — one of VIDEO-CONFIRMED – <file> / PENDING (UI – film next) / N/A (no UI – test-covered + exercised by <journey>)

Video-recording confirmations (§11.4.143 / §11.4.83)

Real-use recordings (Claude-vision analyzed — the HelixAgent ensemble vision is a proven stub, see docs/qa/recording-readiness-20260615/) in /Volumes/T7/Downloads/Recordings/ (project-prefix boba-). Per-row Video cells stay PENDING where a feature has not been *individually* filmed; the recordings below confirm the headline user-facing flows on-screen.

Recording	Surface	Features confirmed	Verdict
boba-cli-orchestrator-demo.mp4	CLI (boba-ctl)	status / health / list — 4 services running & healthy	docs/qa/recordings-20260615/boba-cli-verdict.md
boba-web-search-flow.mp4	Web dashboard	launch → search “debian” → live results	boba-web-search-flow-verdict.md

Recording	Surface	Features confirmed	Verdict
boba-web-feature-tour.mp4 + boba-web-dashboard-tour.mp4	Web dashboard	→ “829 results (288 merged)” complete dashboard load (29 trackers, qBit connected), result rows render qBit + Download buttons (clickable), tab nav (Results/ Active Downloads/ Trackers/ Schedules/ Hooks), Jackett / jackett credentials page, theme • content- script detects a magnet + appends a	boba-web-feature- tour-verdict.md
boba-extension-{scan- detect,popup,options,tour}.mp4	BobaLink extension (MV3, real --load- extension Chromium)	MAGNET badge (non-torrent correctly skipped); popup UI (header, server status, Refresh + Send All, Detected- torrents); options page (7	boba-extension- verdict.md

Recording	Surface	Features confirmed	Verdict
boba-web-tabs-theme-tour.mp4 (+ frames boba-web-tab-active-downloads.png / boba-web-tab-schedules.png / boba-web-tab-hooks.png / boba-web-theme-toggle.png) + boba-web-jackett-add-credential.png	Web dashboard + Jackett page	settings tabs + server config) Active Downloads / Schedules / Hooks tabs each render their empty-state + columns; theme toggle; processing spinner ("Found N results..."); bridge/ qBit- Connected header indicator; Jackett Add- credential dialog (Name/ Username/ Password/ Cookies + Cancel/ Save)	Claude-vision verified 2026-06-16

Honest scope (§11.4.6, no bluff): - Backend services
(download-proxy / merge / boba-jackett / Go) verified **200 server-side inside the VM** and exercised through the web UI recordings. - **Console errors** seen in the web recordings are an **SSH-tunnel SSE/poll artifact** (macOS↔VM), NOT product defects — every endpoint is 200 in the VM. See the feature-tour verdict. - **BobaLink extension: VIDEO-CONFIRMED** — the prior --load-extension tooling gap was closed by building extension/scripts/record-features.mjs (loads the built MV3 artifact into a real Chromium, records the scan/popup/options journeys). Plus its passing test suite. - **TUI / mobile / desktop:** do not exist in this project (server-side + Angular web + browser extension only) — nothing to record.

Component summary

#	Component	Service / Port	Features cataloged
1	Download Proxy + Merge Search Service (Python/FastAPI)	qbittorrent-proxy :7186 / :7187	69
2	qBitTorrent-go backend (Go/Gin)	qbittorrent-proxy-go :7186/:7187/:7188 (opt-in --profile go)	47
3	boba-jackett (Go)	boba-jackett :7189	26
4	Tracker plugins (plugins/*.py)	run inside qbittorrent-proxy	30
5	Angular 21 frontend dashboard (frontend/)	served from :7187	34
6	BobaLink browser extension (extension/)	WXT MV3 client	39
7	WebUI bridge (webui-bridge.py)	host process :7188	4
8	Infrastructure / CLI / shell scripts	host	40

Total features cataloged: 289 (Rev 7: +1 — new GET /api/v1/healthz endpoint)

1. Download Proxy + Merge Search Service (Python/FastAPI) — :7186 / :7187

Entry: download-proxy/src/main.py (starts legacy proxy thread + uvicorn api:app). This is the **shipped default product**.

1a. Merge Search Service — REST API (download-proxy/src/api/routes.py)

Feature	Impl	Wired	Tests	Valid
POST /api/v1/search — async fan-out search, returns search_id (routes.py:283)	implemented	yes	integration tests/integration/test_merge_api.py; e2e test_full_pipeline.py; stress test_merge_search_stress_chaos.py	tested green suite LIVE
POST /api/v1/search/sync — synchronous search (routes.py:377)	implemented	yes	integration test_merge_api.py; stress test_search_stress.py	tested green suite reset tunnel (RW-)
GET /api/v1/search/stream/{id} — SSE result stream (routes.py:547)	implemented	yes	integration test_realtime_streaming.py, test_streaming_browser.py	tested green suite
GET /api/v1/search/{id} — poll search state (routes.py:586)	implemented	yes	integration test_merge_api.py	tested green suite
POST /api/v1/search/{id}/abort — cancel in-flight search (routes.py:619)	implemented	yes	integration test_merge_api.py	tested green suite
GET /api/v1/downloads/active — qBit active torrents (routes.py:629)	implemented	yes	integration test_buttons_api.py	tested green suite
POST /api/v1/auth/qbittorrent — qBit WebUI login (routes.py:671)	implemented	yes	integration test_login_actions.py	tested green suite
	implemented	yes		

Feature	Impl	Wired	Tests	Valid
POST /api/v1/ download — add torrent to qBit (magnet/url) (routes.py:958)			integration test_button_functions.py, test_buttons_api.py; security test_ssrf_and_magnet_auth.py	teste green suite SSRF RW-0
POST /api/v1/ download/upload — upload .torrent file to qBit (routes.py:1123)	implemented	yes	unit tests/unit/api_layer/ test_download_upload.py	unit- integ
POST /api/v1/ download/file — fetch tracker .torrent w/ auth cookies (routes.py:1212)	implemented	yes	integration test_buttons_api.py; security test_ssrf_and_magnet_auth.py	teste green suite SSRF RW-0
POST /api/v1/ magnet — resolve result → magnet link (routes.py:1279)	implemented	yes	integration test_magnet_dialog.py, test_buttons_api.py	teste green suite gap I
GET /api/v1/ theme — read theme state (routes.py:61)	implemented	yes	contract test_crossapp_theme_contract.py	teste green suite
PUT /api/v1/ theme — persist theme (routes.py:67)	implemented	yes	e2e test_theme_runtime.py, test_crossapp_theme.py	teste green suite
GET /api/v1/ theme/stream — SSE theme broadcast (routes.py:76)	implemented	yes	e2e test_crossapp_theme.py	teste green suite

1b. Auth API — per-tracker (download-proxy/src/api/auth.py)

Feature	Impl	Wired	Tests	Validation
GET /auth/ rutracker/ status — RuTracker session	implemented	yes	integration test_auth_and_links.py; e2e test_live_stack_evidence.py	partial — CAPTCHA operator-k (BOB-008)

Feature	Impl	Wired	Tests	Validation
state (auth.py:56)				
GET /auth/ rutracker/ captcha — fetch CAPTCHA image (auth.py:108)	implemented	yes	integration test_auth_and_links.py	partial — operator-k (BOB-008)
POST /auth/ rutracker/ login — username/ password + CAPTCHA login (auth.py:214)	implemented	yes	integration test_auth_and_links.py	partial — operator-k (BOB-008)
				PASS — I 2026-06- RUTRACKER_ cookie pat authentic (2fc29fc); rutracker matrix = 5 results on (docs/qa/s fix- verify-202 §1)
POST /auth/ rutracker/ cookie-login — cookie- based login (auth.py:298)	implemented	yes	integration test_auth_and_links.py	
GET /auth/ nnmclub/ status — NNM-Club session state (auth.py:350)	implemented	yes	e2e test_live_stack_evidence.py:265 (skips — SOURCE→ARTIFACT drift RW-07)	partial — 404 on ru container (RW-07)
POST /auth/ nnmclub/ login — NNM-Club login (auth.py:407)	implemented	yes	e2e test_live_stack_evidence.py	partial — gated
GET /auth/ status — all- tracker aggregate	implemented	yes	integration test_auth_state_ui.py; auth- status cookie-reflection test (§1.1)	PASS — I 2026-06- now reflect RUTRACKER_

Feature	Impl	Wired	Tests	Validation
session state (auth.py:479)				NNMCLUB_CO env before search (9c so dashbo chips show rutracker/ nnmclub g (docs/qa/s fix- verify-202 §3)
POST /auth/ qbittorrent/ logout — qBit WebUI logout (auth.py:532)	implemented	yes	integration test_login_actions.py	tested-gre suite
Kinozal auth (status/ login) REST route	missing	no	none — no kinozal/login route in auth.py	not-implen (search pa cookies vi orchestrat
IPTorrents auth (status/ login) REST route	missing	no	none — no iptorrents/login route in auth.py	not-implen (env creds cookie flow Jackett UI

1c. Search orchestration internals (download-proxy/ src/merge_service/search.py)

Feature	Impl	Wired	Tests
Concurrency caps MAX_CONCURRENT_SEARCHES (429 on saturation)	implemented	yes	concurrency test_orchestrator_semaphore
Per-search tracker semaphore MAX_CONCURRENT_TRACKERS	implemented	yes	concurrency test_tracker

Feature	Impl	Wired	Tests
Public-tracker subprocess fan-out (NDJSON via patched novaprinter)	implemented	yes	e2e test_public_trackers_ret stress test_search_orchestration
Public-tracker deadline kill PUBLIC_TRACKER_DEADLINE_SECONDS	implemented	yes	chaos test_tracker_failu
Private-tracker aiohttp+cookie path (rutracker/kinozal/nmclub/iptorrents)	implemented	yes	integration test_tracker_ (live-gated)
Per-tracker TrackerSearchStat real-time stats	implemented	yes	contract test_tracker_sta property test_tracker_stats_prope
Stderr error classification (_classify_plugin_stderr)	implemented	yes	chaos test_tracker_failu

Feature	Impl	Wired	Tests
TTL-bounded data stores (cachetools) — no leak	implemented	yes	memory test_orchestrator_caches_

1d. Deduplication (download-proxy/src/merge_service/deduplicator.py)

Feature	Impl	Wired	Tests	Validation
Tier 1 dedup — metadata (imdb/tmdb) key	implemented	yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested-green-in-suite
Tier 2 dedup — infohash key	implemented	yes	property test_deduplicator_properties.py	tested-green-in-suite
Tier 3 dedup — name+size key	implemented	yes	property test_deduplicator_properties.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
Tier 4 dedup — fuzzy Levenshtein name match	implemented	yes	property test_deduplicator_properties.py; benchmark test_deduplication_benchmark.py	tested- green-in- suite

1e. Enrichment — per metadata source (download-proxy/src/merge_service/enricher.py)

Feature	Impl	Wired	Tests	Validation
Enrichment: OMDb (enricher.py:113)	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in- suite; ne OMDB_API
Enrichment: TMDB	implemented	yes	stress test_enricher_stress_chaos.py	tested- green; needs TMDB_API
Enrichment: TVMaze	implemented	yes	stress test_enricher_stress_chaos.py	

Feature	Impl	Wired	Tests	Validat
				tested- green-in suite
Enrichment: AniList	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in suite
Enrichment: MusicBrainz	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in suite
Enrichment: OpenLibrary	implemented	yes	stress test_enricher_stress_chaos.py	tested- green-in suite

1f. Tracker validation (download-proxy/src/merge_service/validator.py)

Feature	Impl	Wired	Tests	Validation	Video
Tracker validation HTTP scrape BEP 48 (validator.py:101)	implemented	yes	none dedicated found (covered via merge tests)	not-validated (no dedicated test)	N/A (no UI — exercised by web search flow seeds/leechers column)
Tracker validation UDP scrape BEP 15 (validator.py:162)	implemented	yes	none dedicated found	not-validated (no dedicated test)	N/A (no UI — exercised by web search flow seeds/leechers column)

1g. Hooks (download-proxy/src/api/hooks.py + merge_service/hooks.py)

Feature	Impl	Wired	Tests
GET /api/v1/hooks — list hooks (hooks.py:105)	implemented	yes	unit test_hooks_coverage concurrency test_hooks_race.py; test_hooks_schedules_
POST /api/v1/hooks — create hook (hooks.py:111)	implemented	yes	unit test_hooks_coverage concurrency test_hooks_race.py
DELETE /api/v1/hooks/{id} — delete hook (hooks.py:167)	implemented	yes	unit test_hooks_coverage

Feature	Impl	Wired	Tests
GET /api/v1/hooks/logs — hook execution logs (hooks.py:179)	implemented	yes	unit test_hooks_covera
Hook event dispatch (8 events: search_start..validation_complete) (hooks.py:32)	implemented	yes	concurrency test_hooks_race.py
Hook subprocess execution + sandbox (merge_service/hooks.py)	implemented	yes	security test_no_shell_injecti

1h. Scheduler (download-proxy/src/api/scheduler.py + merge_service/scheduler.py)

Feature	Impl	Wired	Tests	Validat
GET /api/v1/schedules — list schedules (scheduler.py:39)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite; au gap RW
POST /api/v1/schedules — create schedule (scheduler.py:64)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite; au gap RW
GET /api/v1/schedules/{id} — get schedule (scheduler.py:87)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in suite

Feature	Impl	Wired	Tests	Validation
PATCH /api/v1/schedules/{id} — update schedule (scheduler.py:108)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in-suite
DELETE /api/v1/schedules/{id} — delete schedule (scheduler.py:126)	implemented	yes	security test_hooks_schedules_auth.py	tested-green-in-suite
Scheduler engine (load/save/tick driver) (merge_service/scheduler.py)	implemented	yes	integration test_dashboard_automation.py	tested-green-in-suite

1i. App-level routes / health / config / dashboard (download-proxy/src/api/__init__.py)

Feature	Impl	Wired	Tests	Validation
GET /api/v1/healthz — machine-parsable JSON health probe (routes.py:37)	implemented	yes	integration test_merge_api.py	PASS — LIVE 2026-06 — NEW endpoint (137d7ff): bare /healthz w swallowed by the SP catch-all, now mounted under /api/v1 return JSON (\$11.4.43)
GET /health (__init__.py:168)	implemented	yes	integration test_live_containers.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
GET /api/v1/bridge/health (__init__.py:173)	implemented	yes	integration test_bridge_root_liveness.py	tested-green-in-suite
GET /api/v1/config (__init__.py:247)	implemented	yes	integration test_merge_api.py	tested-green-in-suite
GET /api/v1/stats (__init__.py:276)	implemented	yes	observability test_metrics_exist.py	tested-green-in-suite
Jinja2 dashboard GET / (__init__.py:337)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite
Jinja2 dashboard GET /dashboard (__init__.py:342)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite
SPA catch-all GET /{path:path} (__init__.py:347)	implemented	yes	integration test_dashboard_rendering.py	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation

1j. Supporting subsystems (Python)

Feature	Impl	Wired	Tests	Validation
Retry helpers (merge_service/ retry.py)	implemented	yes	covered indirectly	
Jackett autoconfig (Python, merge_service/ jackett_autoconfig.py) — legacy; canonical is boba-jackett	implemented	yes	contract test_jackett_autoconfig_contract.py e2e test_jackett_autoconfig_e2e.py benchmark test_jackett_autoconfig_perf.py	
Credential scrubbing log filter (config/ log_filter.py)	implemented	yes	security test_credential_scrubbing.py	

1k. Legacy download proxy (plugins/download_proxy.py :7186)

Feature	Impl	Wired	Tests	Validation
HTTP proxy → qBittorrent WebUI w/ tracker URL interception (DownloadHandler :767)	implemented	yes	integration test_bridge_header_rewrite.py	tested green suite
Theme injection into proxied	implemented	yes	integration test_bridge_theme_injection.py	tested green suite

Feature	Impl	Wired	Tests	Valid
WebUI (plugins/ theme_injector.py)				

2. qBitTorrent-go backend (Go/Gin) — opt-in --profile go

Entry: qBitTorrent-go/cmd/qbittorrent-proxy/main.go (routes registered :54-101). **Skeleton/rewrite-in-progress** (per AGENTS.md + docs/migration/PARITY_GAPS.md). NOT running by default.

2a. HTTP handlers (internal/api/*.go)

Feature	Impl	Wired	Tests	Valid
GET /health — HealthHandler (api/ health.go:5)	implemented	opt-in	go internal/api/ api_test.go	unit- (go t
GET /api/v1/config — ConfigHandler (api/ download.go:210)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/bridge/ health — BridgeHealthHandler (api/download.go:189)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/search — SearchHandler (api/ search.go:14)	implemented (proxies qBit search; no plugin fan- out)	opt-in	go api_test.go, coverage_test.go	unit-

Feature	Impl	Wired	Tests	Valid
POST /api/v1/search/ sync — SearchSyncHandler (api/search.go:50)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/search/ stream/:id — SearchStreamHandler SSE (api/ search.go:91)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/search/:id — GetSearchHandler (api/search.go:144)	implemented	opt-in	go getsearch_results_test.go, coverage_test.go	unit-
POST /api/v1/ search/:id/abort — AbortSearchHandler (api/search.go:177)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/download — DownloadHandler (api/download.go:17)	partial — returns mock success, no qBit integration	opt-in	go coverage_test.go	unit- func hole
POST /api/v1/download/ file — DownloadFileHandler (api/download.go:41)	implemented	opt-in	go coverage_test.go	unit-
POST /api/v1/magnet — MagnetHandler (api/ download.go:86)	implemented	opt-in	go api_test.go, magnet_stress_chaos_test.go	unit stres chao
GET /api/v1/downloads/ active — ActiveDownloadsHandler (api/download.go:147)	stub — returns empty	opt-in	go coverage_test.go	unit- func hole

Feature	Impl	Wired	Tests	Valid
	array, no qBit query			
POST /api/v1/auth/ qbittorrent — QBittorrentAuthHandler (api/download.go:156)	implemented	opt-in	go coverage_test.go	unit-
GET /api/v1/theme — GetThemeHandler (api/ theme.go:90)	implemented	opt-in	go theme_hardening_test.go, coverage_test.go	unit-
PUT /api/v1/theme — PutThemeHandler (api/ theme.go:96)	implemented	opt-in	go theme_hardening_test.go	unit-
GET /api/v1/hooks — ListHooksHandler (api/ hooks.go:91)	implemented (store only, no dispatch)	opt-in	go api_test.go, coverage_test.go	unit-
POST /api/v1/hooks — CreateHookHandler (api/hooks.go:97)	implemented (store only)	opt-in	go api_test.go	unit-
DELETE /api/v1/ hooks/:id — DeleteHookHandler (api/hooks.go:109)	implemented (store only)	opt-in	go api_test.go	unit-
GET /api/v1/schedules — ListSchedulesHandler (api/ scheduler_api.go:86)	implemented (store only)	opt-in	go coverage_test.go	unit- no d loop (RW sche neve

Feature	Impl	Wired	Tests	Validation	Video
POST /api/v1/schedules — CreateScheduleHandler (api/scheduler_api.go:92)	implemented (store only)	opt-in	go coverage_test.go		unit- never (RW)
DELETE /api/v1/schedules/:id — DeleteScheduleHandler (api/scheduler_api.go:104)	implemented (store only)	opt-in	go coverage_test.go		unit-

2b. qBittorrent Web API client (internal/client/*.go)

Feature	Impl	Wired	Tests	Validation	Video
NewClient — construct + login (client/client.go:18)	implemented	opt-in	go client/client_test.go	unit-only	N/A (no UI — go-test-covered client method; opt-in profile)
Login — POST /api/v2/auth/login, store SID (client/auth.go:11)	implemented	opt-in	go client/auth_test.go	unit-only	N/A (no UI — go-test-covered client method; opt-in profile)
IsAuthenticated — SID presence (client/auth.go:43)	implemented	opt-in	go client/auth_test.go	unit-only	N/A (no UI — go-test-covered client method; opt-in profile)
GetSID — thread-safe SID accessor	implemented	opt-in	go client/client_test.go	unit-only	N/A (no UI — go-test-covered

Feature	Impl	Wired	Tests	Validation	Video
(client/ client.go:39)					client method; opt-in profile)
GetTorrents — GET /api/v2/ torrents/info (client/ torrents.go:13)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
AddTorrent — POST /api/v2/ torrents/add URL (client/ torrents.go:30)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
AddTorrentFile — POST multipart /api/ v2/torrents/add (client/ torrents.go:54)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
GetAppVersion — GET /api/v2/ app/version (client/ torrents.go:87)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
StartSearch — POST /api/v2/ search/start (client/ search.go:27)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
GetSearchResults — GET /api/v2/ search/results	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered

Feature	Impl	Wired	Tests	Validation	Video
(client/ search.go:55)					client method; opt-in profile)
StopSearch — POST /api/v2/ search/stop (client/ search.go:84)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
SearchStatus — GET /api/v2/ search/status (client/ search.go:100)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)
ListPlugins — GET /api/v2/ search/plugins (client/ search.go:121)	implemented	opt-in	go client/ coverage_test.go	unit-only	N/A (no UI — go-test- covered client method; opt-in profile)

2c. Services (internal/service/*.go)

Feature	Impl	Wired	Tests
Merge search orchestrator RunSearch poll/dedup loop (service/ merge_search.go:183)	implemented (qBit search only; no python-style plugin fan-out)	opt-in	go service/service_test.go, runsearch_dedup_test.go
StartSearch / GetSearchStatus / AbortSearch lifecycle (service/ merge_search.go:101-143)	implemented	opt-in	go getsearchstatus_hardening_test.go
Dedup key (FileURL→name+size) (service/ merge_search.go:269)	implemented	opt-in	go runsearch_dedup_test.go

Feature	Impl	Wired	Tests
FetchTorrent tracker fetch (service/ merge_search.go:276)	stub — returns “not yet implemented”	opt-in	none
Stats counters (service/ merge_search.go:280)	implemented	opt-in	go coverage_test.go
SSE broker pub/sub (service/sse_broker.go)	implemented (defined but not wired — handlers SSE inline)	opt-in	none dedicated
Metadata enricher (Go)	missing (RW-11)	no	none
Scheduler driver loop (Go)	missing (RW-10) — CRUD only, no cron execution	no	none

2d. webui-bridge binary (Go) + cross-cutting

Feature	Impl	Wired	Tests	Validation	Video
webui- bridge binary (cmd/ webui- bridge/ main.go)	partial (health + root only)	opt-in	none dedicated	not- validated	N/A (no UI of its own - proxy binary; opt in profile)
CORS + logging middleware (internal/ middleware/)	implemented	opt-in	go middleware/ middleware_test.go	unit-only	N/A (no UI — go-test- covered middlewar opt-in profile)
Models / data types (internal/ models/)	implemented	opt-in	go models/ models_test.go	unit-only	N/A (no UI — go-test- covered data types, opt-in profile)
Env config loader	implemented	opt-in	go config/ config_test.go	unit-only	N/A (no UI — go-test- covered

Feature	Impl	Wired	Tests	Validation	Video
(internal/ config/)					config loader; opt in profile)
Log redactor (internal/ logging/ redactor.go)	implemented	opt-in	go logging/ redactor_test.go	unit-only	N/A (no UI — go-test- covered redactor; opt-in profile)

3. boba-jackett (Go) — :7189

Entry: qBittorrent-go/cmd/boba-jackett/main.go; router internal/jackettapi/router.go. Owns Jackett credentials + indexer overrides + autoconfig run history; encrypted SQLite at /config/boba.db.

3a. HTTP endpoints (internal/jackettapi/router.go + handlers)

Feature	Impl	Wired	Tests	Validation
GET /healthz (router.go:60, health.go)	implemented	yes	go jackettapi/ health_test.go	unit (go)
GET /openapi.json (router.go:64, openapi.go)	implemented	yes	go jackettapi/ openapi_test.go; contract tests/contract/ openapi_test.go	unit + contract (go)
GET /api/v1/ jackett/ credentials — list (router.go:67, credentials.go)	implemented	yes	go credentials_test.go; integration tests/ integration/ jackett_db_test.go	unit + integrati (go)
POST /api/v1/ jackett/ credentials — create/update (router.go:67, credentials.go)	implemented	yes	go credentials_test.go	unit + integrati (go)

Feature	Impl	Wired	Tests	Validation
DELETE /api/v1/jackett/credentials/{name} (router.go:77, credentials.go)	implemented	yes	go credentials_test.go	unit + integrati (go)
GET /api/v1/jackett/indexers — list configured (router.go:86, indexers.go)	implemented	yes	go jackettapi/indexers_test.go	unit (go)
POST/PATCH/DELETE /api/v1/jackett/indexers/{id} (router.go:93, indexers.go)	implemented	yes	go indexers_test.go	unit (go)
POST /api/v1/jackett/indexers/{id}/test — test config (indexers.go:239)	implemented	yes	go indexers_test.go	unit (go)
GET /api/v1/jackett/catalog — indexer catalog (router.go:112, catalog.go)	implemented	yes	go jackettapi/catalog_test.go; benchmark catalog_bench_test.go	unit + bench (g
POST /api/v1/jackett/catalog/refresh (router.go:119, catalog.go)	implemented	yes	go catalog_test.go	unit (go)
GET /api/v1/jackett/	implemented	yes	go jackettapi/runs_test.go; e2e tests/	unit + e2 (go)

Feature	Impl	Wired	Tests	Validation
autoconfig/runs — run history (router.go:137, runs.go)			e2e/ jackett_management_test.go	
GET /api/v1/jackett/ autoconfig/runs/{id} — run detail (router.go:144, runs.go)	implemented	yes	go runs_test.go	unit (go)
POST /api/v1/jackett/ autoconfig/run — trigger run (router.go:151, runs.go)	implemented	yes	go runs_test.go; e2e tests/e2e/ jackett_management_test.go	unit + e2e (go)
GET/POST /api/v1/jackett/overrides (router.go:160, overrides.go)	implemented	yes	go jackettapi/ overrides_test.go	unit (go)
DELETE /api/v1/jackett/overrides/{env} (router.go:170, overrides.go)	implemented	yes	go overrides_test.go	unit (go)
Admin auth middleware (admin/admin on mutating routes) (jackettapi/auth_middleware.go)	implemented	yes	go auth_middleware_test.go	unit (go)
Hardened CORS middleware (jackettapi/cors_middleware.go)	implemented	yes	go cors_middleware_test.go	unit (go)

3b. Autoconfig engine + persistence (internal/jackett/, internal/db/)

Feature	Impl	Wired	Tests	Validation	Vi
Jackett autoconfig engine	implemented	yes	go jackett/ autoconfig_test.go, autoconfig_bench_test.go	unit + bench (go)	N/ — cov en

Feature	Impl	Wired	Tests	Validation	View
(jackett/ autoconfig.go)					ex by jac au run
Fuzzy indexer matcher (jackett/ matcher.go)	implemented	yes	go jackett/ matcher_test.go	unit (go)	N/ — cov ma
Jackett HTTP client (jackett/ client.go)	implemented	yes	go jackett/ client_test.go	unit (go)	N/ — cov cli
Encrypted SQLite — AES-256- GCM crypto (db/crypto.go)	implemented	yes	go db/crypto_test.go; security tests/security/ credential_leak_test.go	unit + security (go)	N/ — + s cov cry
SQLite migrations (db/ migrate.go)	implemented	yes	go db/migrate_test.go	unit (go)	N/ — cov mi
Credential / indexer / runs repos (db/repos/ *.go)	implemented	yes	go db/repos/*_test.go	unit (go)	N/ — cov rep
.env bootstrap import (bootstrap/ *.go)	implemented	yes	go bootstrap/ bootstrap_test.go	unit (go)	N/ — cov bo
Master-key ensure (first- boot) (bootstrap/ *.go)	implemented	yes	go bootstrap/ bootstrap_test.go	unit (go)	N/ — cov fir key en
.env file parse/write (envfile/*.go)	implemented	yes	go envfile/ parse_test.go, write_test.go	unit (go)	N/ — cov en O)

4. Tracker plugins (plugins/*.py)

Plugin contract: class with url, name, supported_categories, search(), download_torrent(). Installed into container by install-plugin.sh. The install-plugin.sh PLUGINS array names **44 plugins** (curated install set); the plugins/ tree carries **27 *.py files** (24 tracker/aggregator plugins + 3 support: nova2, novaprinter, socks, helpers, env_loader, theme_injector, download_proxy). Many curated names have NO matching file in this repo tree (the install script fetches/ expects them elsewhere) — flagged honestly below.

4a. Plugins present in plugins/ AND named in the curated array

Plugin	Type	Impl	Wired (curated?)	Tests
eztv.py	public	implemented	yes (curated)	stress test_plugin_p
limetorrents.py	public	implemented	yes (curated)	stress test_plugin_p multi-word tests/un. test_plugin_multiwor stress/chaos tests/s test_multiword_query (\$11.4.85)
piratebay.py	public	implemented	yes (curated)	stress test_plugin_p
solidtorrents.py	public	implemented	yes (curated)	stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
torlock.py	public	implemented	yes (curated)	stress test_plugin_p
rutor.py	public (anonymous)	implemented	yes (curated)	stress test_plugin_p
nyaa.py	public	implemented	yes (curated)	stress test_plugin_p
anilibra.py	public	implemented	yes (curated)	stress test_plugin_p
bitsearch.py	public	implemented	yes (curated)	stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
gamestorrents.py	public	implemented	yes (curated)	stress test_plugin_p
kickass.py	public	implemented (degrades to honest empty)	yes (curated)	stress test_plugin_p
megapeer.py	public	implemented	yes (curated)	stress test_plugin_p
tokyotoshokan.py	public	implemented	yes (curated)	stress test_plugin_p

Plugin	Type	Impl	Wired (curated?)	Tests
torrentgalaxy.py	public	implemented	yes (curated)	stress test_plugin_p
torrentkitty.py	public	implemented	yes (curated)	stress test_plugin_p
yts.py	public	implemented	yes (curated)	stress test_plugin_p
jackett.py	aggregator	implemented	yes (curated)	covered via jackett a
iptorrents.py	private (freeleech- only)	implemented	yes (curated)	integration tests/in test_iporrents.py
rutracker.py		implemented	yes (curated)	integration test_tra injection tests/unit/

Plugin	Type	Impl	Wired (curated?)	Tests
	private (cookies / CAPTCHA)			(§1.1); stress test_plugin_parsers_ §11.4.85)
kinozal.py	private	implemented	yes (curated)	integration test_tra
nmclub.py	private (cookies)	implemented	yes (curated)	integration test_tra

4b. Plugin support modules

Module	Type	Impl	Wired	Tests	Valida
nova2.py — nova3 search harness	support	implemented	yes	covered via search e2e	tested- green-i suite

Module	Type	Impl	Wired	Tests	Valida
novaprinter.py — result emitter (NDJSON- patched)	support	implemented	yes	covered via search e2e	tested- green-i suite
helpers.py — HTTP/retrieve helpers	support	implemented	yes	covered via search e2e	tested- green-i suite
socks.py — SOCKS proxy support	support	implemented	yes	covered indirectly	not- validat
env_loader.py — env/credential loader for plugins	support	implemented	yes	covered via tracker auth tests	tested- green-i suite
Per-plugin download_torrent() path (.torrent fetch)	feature	implemented	yes	integration test_buttons_api.py (download/file)	tested- green-i suite
Freeleech-only download policy (IPTorrents [free] tag)	policy	implemented	yes	integration test_iptorrents.py	tested- green-i suite

Module	Type	Impl	Wired	Tests	Validation
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4c. Discrepancy rows (honest, §11.4.6)

Plugin	Type	Impl	Wired (curated?)	Tests	Validation	Vi
torrentproject / torrentscsv	public (curated names)	named in install- plugin.sh PLUGINS array but NO matching plugins/ *.py file in tree	curated- name-only	none	not- implemented- in-repo-tree (install script expects them elsewhere)	N/ im in — na plu
23 other curated names absent from tree: academictorrents, ali213, audiobookbay, bitru, bt4g, btsow, extratorrent, glotorrents, linuxtracker, one337x, pctorrent, pirateiro, rockbox, snowfl, therarbg, torrentdownload, torrentfunk, xfsub, yihua, yourbittorrent (+ torrentproject, torrentscsv)	public (curated names)	named in curated array; NO plugins/ *.py in this tree	curated- name-only	none	not- implemented- in-repo-tree (name in install array; file absent)	N/ im in — na plu

NOTE (§11.4.6, CORRECTION vs Rev 2): the install-plugin.sh PLUGINS array contains **44 entries**, not 12. Of those, ~21 have a matching plugins/*.py file in this repo; the rest are curated names with no file present in the tree. Stated as fact, not asserted as working.

5. Angular 21 frontend dashboard (frontend/)

Standalone SPA, signals-based; built to download-proxy/src/ui/dist/frontend, served from :7187. Vitest unit tests; Playwright e2e under frontend/e2e.

5a. Dashboard component controls (components/dashboard/dashboard.component.ts)

Feature	Impl	Wired	Tests
Search box (query signal) + submit	implemented	yes	vitest dashboard.component.spec.ts; py integration test_dashboard_rendering.py, test_ui_comprehensive.py
Result grid — sortable (name/type/size/seeds/leechers/quality/sources)	implemented	yes	vitest dashboard.component.spec.ts
Results tab	implemented	yes	py test_ui_comprehensive.py
Active Downloads tab + view (dashboard.component.ts:745 loadDownloads)	implemented	yes	py test_buttons_api.py
Trackers tab + per-tracker auth chips (expandable)	implemented	yes	py test_auth_state_ui.py

Feature	Impl	Wired	Tests
Schedules tab	implemented	yes	py test_dashboard_automation.py
Hooks tab	implemented	yes	py test_dashboard_automation.py
Per-row Magnet button → magnet dialog trigger	implemented	yes	vitest magnet-dialog.component.spec.ts; py test_magnet_dialog.py
Per-row add-to-qBittorrent (qBit) button	implemented	yes	py test_button_functions.py, test_buttons_api.py
Per-row Download (.torrent) button	implemented	yes	py test_buttons_api.py
Per-row processing spinner (isBusy())	implemented	yes	vitest dashboard.component.spec.ts
Bridge health indicator + retry button	implemented	yes	py test_bridge_root_liveness.py

Feature	Impl	Wired	Tests
Tracker-stat dialog trigger	implemented	yes	py contract test_tracker_stats_contract.py

5b. Dialog + UI components

Feature	Impl	Wired	Tests	V
Magnet dialog — readonly textarea + Copy/Open/Add/Close (magnet- dialog.component.ts)	implemented	yes	vitest magnet- dialog.component.spec.ts; py test_magnet_dialog.py	t g s
qBit login dialog — user/pass/remember/ Login/Cancel (qbit- login- dialog.component.ts)	implemented	yes	vitest qbit-login- dialog.component.spec.ts; py test_login_actions.py, test_auth_state_ui.py	t g s
Theme picker — palette dropdown + light/dark toggle (theme- picker.component.ts)	implemented	yes	vitest theme- picker.component.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	t g s
Confirm dialog — title/ message/Confirm/ Cancel/backdrop (confirm- dialog.component.ts)	implemented	yes	vitest confirm- dialog.component.spec.ts	u (
Tracker-stat dialog — JSON view + Copy + status badge + notes table + Esc-close (tracker-stat- dialog.component.ts)	implemented	yes	vitest tracker-stat- dialog.component.spec.ts; py contract test_tracker_stats_contract.py	t g s
Toast container — info/success/warning/ error auto-dismiss	implemented	yes	vitest toast- container.component.spec.ts	u (

Feature	Impl	Wired	Tests	Validation
(toast-container.component.ts)				
Site footer (site-footer.component.ts)	implemented	yes	vitest	unit (vitest)

5c. Services (frontend/src/app/services/)

Feature	Impl	Wired	Tests	Validation
API service — HTTP wrapper (search/download/magnet/auth/schedules/hooks/downloads/stats/config), 15s timeout, runtime baseUrl (api.service.ts)	implemented	yes	vitest api.service.spec.ts	unit (vitest)
SSE service — EventSource live result streaming + bearer-token (sse.service.ts)	implemented	yes	vitest sse.service.spec.ts; py integration test_realtime_streaming.py, test_streaming_browser.py	tested-green-in-suite
Theme service — palette/mode + localStorage + cross-app sync (theme.service.ts)	implemented	yes	vitest theme.service.spec.ts; py e2e test_theme_runtime.py, test_crossapp_theme.py	tested-green-in-suite
Toast service — toast queue state (toast.service.ts)	implemented	yes	vitest toast.service.spec.ts	unit (vitest)
Dialog service — promise-based confirm state (dialog.service.ts)	implemented	yes	vitest dialog.service.spec.ts	unit (vitest)

5d. Jackett page (frontend/src/app/jackett/)

Feature	Impl	Wired	Tests
Credentials — list + Add/Edit/Delete (per-row, window.confirm) + banners (jackett/credentials/credentials.component.ts)	implemented	yes	vitest credentials.component.spec.ts
Credential edit dialog — name/username/password/cookies/Save/Cancel (credential-edit-dialog.component.ts)	implemented	yes	vitest credential-edit-dialog.component.spec.ts
Indexers — Configured/Catalog/History tab container (jackett/indexers/indexers.component.ts)	implemented	yes	vitest indexers.component.spec.ts
Indexers — Configured tab (list + per-row remove)	implemented	yes	vitest indexers.component.spec.ts
Indexers — Catalog tab (list + per-row add)	implemented	yes	vitest indexers.component.spec.ts
Indexers — History tab (autoconfig run history)	implemented	yes	vitest indexers.component.spec.ts
Indexer add dialog — credential dropdown + Save/Cancel + error banner (indexer-add-dialog.component.ts)	implemented	yes	vitest indexer-add-dialog.component.spec.ts
IPTorrents cookie-flow component (iptorrents-cookie-flow.component.ts)	implemented	yes	vitest (jackett specs)

5e. Frontend infra

Feature	Impl	Wired	Tests	Validation	Video
Frontend coverage thresholds (40% lines/branches/funcs/stmts, v8)	infra	yes	vitest.config.ts	enforced in vitest config	N/A (no UI — build/coverage infra)

6. BobaLink browser extension (extension/)

WXT + TypeScript Manifest-V3. Detects magnet/.torrent links and forwards to merge service :7187. Status detail: docs/browser_extension/Status.md. Built zips at extension/.output/bobalink-1.0.0-{chrome,firefox}.zip.

6a. User-facing surfaces

Feature	Impl	Wired	Tests	Validation	Video
Popup — per-row Send button (src/popup/, entrypoints/popup.ts)	implemented	yes	a11y tests/a11y/popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Send-All button	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Refresh button + status indicator	implemented	yes	a11y popup.a11y.test.ts	tested-green-in-suite	VIDE CONF 2026- boba-e popup.

Feature	Impl	Wired	Tests	Validation	Video
(online/ offline/ warning)					docs/q record boba-e verdic
Popup — torrent list (name + type + infohash prefix + sent badge)	implemented	yes	a11y popup.a11y.test.ts	tested- green-in- suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Popup — Open Options link	implemented	yes	a11y popup.a11y.test.ts	tested- green-in- suite	VIDE CONF 2026- boba-e popup. docs/q record boba-e verdic
Options page (src/options/ options.ts)	implemented	yes	a11y options.a11y.test.ts; integration options- background.test.ts	tested- green-in- suite	VIDE CONF 2026- boba-e option docs/q record boba-e verdic
Content script — link/ text highlight overlay (src/ content/ highlight.ts, entrypoints/ content.ts)	implemented	yes	integration content- background.test.ts; security content- xss.test.ts	tested- green-in- suite	VIDE CONF 2026- boba-e detect docs/q record boba-e verdic
Background service worker — message hub (src/ background/ entrypoints/ background.ts)	implemented	yes	integration tests/ integration/*- background.test.ts	tested- green-in- suite	N/A (n covere exerci extens popup

Feature	Impl	Wired	Tests	Validation	Video
Context-menu entries (right-click torrent actions)	implemented	yes	integration *-background.test.ts	tested-green-in-suite	N/A (no coverage exercised by extension popup)
Keyboard shortcuts (scan-now / highlight-toggle / send-all)	implemented	yes	integration *-background.test.ts	tested-green-in-suite	N/A (no coverage exercised by extension popup)
Action badge (detected count) + notifications	implemented	yes	integration *-background.test.ts	tested-green-in-suite	N/A (no coverage exercised by extension popup)

6b. Scanners (src/scanner/)

Feature	Impl	Wired	Tests	Validation	Video
Base scanner — computeStableId() lifecycle (scanner/base.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A (no UI test coverage; scanner exercised by extension scanner detected)
Link scanner — magnet/.torrent (scanner/link-scanner.ts)	implemented	yes	security scanner-hostile-input.test.ts; perf scanner.perf.test.ts	tested-green-in-suite	N/A (no UI test coverage; scanner exercised by extension scanner detected)
Text scanner — bare magnet text (scanner/text-scanner.ts)	implemented	yes	security scanner-hostile-input.test.ts	tested-green-in-suite	N/A (no UI test coverage; scanner exercised by

Feature	Impl	Wired	Tests	Validation	Video
Scanner orchestrator — dedup + mutation-observer re-scan (scanner/orchestrator.ts)	implemented	yes	perf orchestrator-scaling.perf.test.ts	tested-green-in-suite	extended scan detected N/A (no UI — test-covered scanner; exercised by extension scan-detect)
Site DB — per-site selector match rules (scanner/site-db.ts)	implemented	yes	vitest unit	tested-green-in-suite	extended scan detected N/A (no UI — test-covered scanner; exercised by extension scan-detect)

6c. Parsers (src/parser/)

Feature	Impl	Wired	Tests	Validation	Video
Bencode parser (parser/bencode.ts)	implemented	yes	security bencode-torrentfile-hostile.test.ts; perf parsers.perf.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)
Magnet parser (parser/magnet.ts)	implemented	yes	security infohash-detection-hostile.test.ts; perf magnet.perf.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)

Feature	Impl	Wired	Tests	Validation	Video
Torrent-file parser (parser/torrent-file.ts)	implemented	yes	security bencode-torrentfile-hostile.test.ts	tested-green-in-suite	N/A (no UI — test-covered parser; exercised by extension scan-detect)

6d. API client + offline (src/api/)

Feature	Impl	Wired	Tests	Validation	Video
Boba API client — POST /api/v1/download, retry + rate-limit (api/boba-client.ts)	implemented	yes	vitest unit; chaos tests/chaos/boba-client-resilience.chaos.test.ts	tested-green-in-suite	N/A (no UI — test-covered API client; exercised by extension Ser
Health probe — probeHealth() (api/health.ts)	implemented	yes	vitest unit	tested-green-in-suite	N/A (no UI — test-covered health probe; exercised by extension stat ind
Offline send queue — IndexedDB enqueue + re-dispatch (api/queue.ts)	implemented	yes	chaos tests/chaos/queue.chaos.test.ts	tested-green-in-suite	N/A (no UI — test-covered offline queue cha cov

6e. Shared utilities + tabgroups (src/shared/, src/tabgroups/)

Feature	Impl	Wired	Tests	Validation	Video
Crypto — AES-256-GCM token encrypt/decrypt (shared/crypto.ts)	implemented	yes	security crypto- tamper.test.ts, no-hardcoded- secret.test.ts	tested- green-in- suite	N/A (no UI — covered crypto- exerci by extens Send token path)
Storage — namespaced chrome.storage.local wrapper (shared/ storage.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — covered storage wrapp
Constants — site selectors / rate- limit / retry config (shared/ constants.ts)	implemented	yes	covered via scanner tests	tested- green-in- suite	N/A (no UI — covered consta
Errors — Network/ Server/Storage error types (shared/ errors.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — covered error types)
Events — TypedEventEmitter bus (shared/ events.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — covered event bus)
Logger — namespaced, no- secret (shared/ logger.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — covered logger
Utils — debounce / TokenBucket rate limiter (shared/ utils.ts)	implemented	yes	vitest unit	tested- green-in- suite	N/A (no UI — covered utils/r limite
Tab groups — batch grouped-tab torrents (tabgroups/ index.ts)	implemented	yes	vitest unit (tab-group Challenge per ext Status)	tested- green-in- suite	N/A (no UI of own — test- cover

Feature	Impl	Wired	Tests	Validation	Video
					tab-gr batch exerci by exten: Send-

6f. i18n locales (extension/src/public/_locales/)

Feature	Impl	Wired	Tests	Validation	Video
Locale en (English)	implemented	yes	i18n tests/ i18n/locale-safety.test.ts	tested-green-in-suite	N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options)
Locale de (German)	implemented	yes	i18n locale-safety.test.ts	tested-green-in-suite	N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options)
Locale es (Spanish)	implemented	yes	i18n locale-safety.test.ts	tested-green-in-suite	N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension

Feature	Impl	Wired	Tests	Validation	Video
Locale fr (French)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options) N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale it (Italian)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale ja (Japanese)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered; surfaced inside extension popup/ options)
Locale pt (Portuguese)	implemented	yes	i18n locale- safety.test.ts	tested- green-in- suite	N/A (no UI of its own — i18n locale data; test- covered;

Feature	Impl	Wired	Tests	Validation	Video
Locale ru (Russian)	implemented	yes	i18n locale-safety.test.ts	tested-green-in-suite	surfaced inside extension popup/options) N/A (no UI of its own — i18n locale data; test-covered; surfaced inside extension popup/options)

6g. Live round-trip

Feature	Impl	Wired	Tests	Validation
Live download endpoint round-trip (tests/live/download-endpoint.live.test.ts)	implemented	yes (live-gated)	live download-endpoint.live.test.ts	partial — operator-gated live env

7. WebUI bridge (webui-bridge.py) — host process :7188

Bridges qBittorrent WebUI with private-tracker auth. NOT a container.

Feature	Impl	Wired	Tests	Valid
WebUI proxy + header rewrite (webui-bridge.py)	implemented	host	integration test_webui_bridge.py, test_bridge_header_rewrite.py; stress test_bridge_stress_chaos.py	tested green suite

Feature	Impl	Wired	Tests	Valid
Private-tracker auth bridging (rutracker/ kinozal/nnmclub/ iptorrents)	implemented	host	integration test_webui_bridge.py	partial live-g
Theme injection into bridged WebUI (plugins/ theme_injector.py)	implemented	host	integration test_bridge_theme_injection.py	tested green suite
Bridge root liveness (/)	implemented	host	integration test_bridge_root_liveness.py	tested green suite

8. Infrastructure / CLI / shell scripts

8a. boba-ctl CLI orchestrator (Go: cmd/boba-ctl/ main.go + wrapper scripts/boba-ctl.sh)

Feature	Impl	Wired	Tests	Validation	Video
boba-ctl up — start stack (cmd/boba-ctl/ main.go:32)	implemented	host	go cmd/boba-ctl/ projectroot_test.go	tested- green-in- suite	N/A (C covere by bob demo, orchest demo.mp
boba-ctl down — stop stack (main.go:34)	implemented	host	go cmd/boba-ctl/ projectroot_test.go	tested- green-in- suite	N/A (C covere by bob demo,

Feature	Impl	Wired	Tests	Validation	Video
boba-ctl status — service status (main.go:36)	implemented	host	go cmd/boba-ctl/ projectroot_test.go	tested- green-in- suite + LIVE	orchest demo.mp VIDEO CONF 2026- boba-cl orchest demo.mp docs/qa recordi boba-cl
boba-ctl health — health probe (main.go:38, cmd/boba-ctl/ health_test.go)	implemented	host	go cmd/boba-ctl/ health_test.go	tested- green-in- suite + LIVE	VIDEO CONF 2026- boba-cl orchest demo.mp
boba-ctl list — list services (main.go:40)	implemented	host	go cmd/boba-ctl/ projectroot_test.go	tested- green-in- suite + LIVE	VIDEO CONF 2026- boba-cl orchest demo.mp
boba-ctl.sh wrapper — compose- compat shim (up/down/ps/ config/pull + passthrough) (scripts/boba- ctl.sh)	implemented	host	covered via boba- ctl go tests	tested- green-in- suite	N/A (C test-co exercis ctl CLI

8b. Lifecycle scripts (repo root)

Feature	Impl	Wired	Tests	Valid
start.sh — container bring-up + Jackett key extraction/ injection	implemented	host	covered via e2e test_fixtures_bring_up_services.py	tested green suite
stop.sh — stop/remove/ purge	implemented	host	none dedicated	not- valida

Feature	Impl	Wired	Tests	Valid
setup.sh — one-time setup	implemented	host	none dedicated	not-validated
start-proxy.sh — proxy container entry helper	implemented	host	none dedicated	not-validated
install-plugin.sh — copy curated plugins → container engines	implemented	host	covered indirectly (e2e search)	tested green suite
init-qbit-password.sh — qBit admin password bootstrap	implemented	host	none dedicated	not-validated
fix-qbit-password.sh — qBit admin password repair	implemented	host	none dedicated	not-validated
setup-webui-bridge-service.sh + webui-bridge.service — systemd unit for bridge	implemented	host	none dedicated	not-validated

8c. Test / CI wrappers (repo root)

Feature	Impl	Wired	Tests	Validation
ci.sh — manual CI (syntax+unit+integration+e2e+health)	implemented	host	self (is the gate)	tested-green-in-suite

Feature	Impl	Wired	Tests	Validation
run-all-tests.sh — full suite (hardcoded podman)	implemented	host	self	partial — fails on docker- only host (known gotcha)
test.sh — quick validation wrapper	implemented	host	self	tested- green-in- suite
test-all.sh — full validation wrapper	implemented	host	self	tested- green-in- suite
test-full.sh — extended validation wrapper	implemented	host	self	tested- green-in- suite

8d. scripts/ helpers

Feature	Impl	Wired	Tests	Validation
scripts/add-submodules.sh — submodule bootstrap	implemented	host	none dedicated	not- validated
scripts/audit-plugins.sh — plugin discrepancy audit	implemented	host	none dedicated	not- validated
	implemented	host		

Feature	Impl	Wired	Tests	Validation
scripts/build-releases.sh — build extension/Go release artifacts			none dedicated	not-validated
scripts/codegraph_validate.sh — CodeGraph index validation (§11.4.78)	implemented	host	self	tested-green-in-suite
scripts/workable-items-export.sh — docs regeneration wrapper (§11.4.93/§11.4.65; renamed 2026-08-15 BOB-104 from docs_chain.sh — that name was a misnomer)	implemented	host	self	tested-green-in-suite
constitution/submodules/docs_chain/docs_chain — real §11.4.106 Docs Chain engine (Go binary, pinned to helixcode-v1.1.0)	implemented	host	self	tested-green-in-suite
scripts/ensure-macos-tunnel.sh — 0.0.0.0 SSH tunnel	implemented	host	none dedicated (log qa-results/tunnel-keepalive.log)	not-validated; LAN threat-model RW-05
scripts/tunnel-keepalive.sh — tunnel self-heal	implemented	host	none dedicated	not-validated
scripts/freeze-openapi.sh — freeze boba-jackett OpenAPI snapshot	implemented	host	contract tests/contract/openapi_test.go	tested-green-in-suite
scripts/generate_markdown_exports.sh — md→html/pdf export (§11.4.65)	implemented	host	self	tested-green-in-suite
scripts/helixqa.sh — HelixQA harness driver	implemented	host	self	not-validated

Feature	Impl	Wired	Tests	Validation
scripts/opencode-helixqa.sh — OpenCode HelixQA driver	implemented	host	self	not-validated
scripts/install_git_hooks.sh — git hook installer (§11.4.75)	implemented	host	self	tested-green-in-suite
scripts/ pre_build_verification.sh — pre-build gate sweep	implemented	host	self (is the gate)	tested-green-in-suite
scripts/pre_code_review.sh — code-review gate (§11.4.125)	implemented	host	self	tested-green-in-suite
scripts/ run_all_challenges.sh — challenge bank runner	implemented	host	self	not-validated
scripts/run-tests.sh — test runner helper	implemented	host	self	tested-green-in-suite
scripts/scan.sh — repo scan helper	implemented	host	none dedicated	not-validated
scripts/track-build-resources.sh — build resource sampler (§11.4.24)	implemented	host	none dedicated	not-validated

8e. Infra / compose / startup

Feature	Impl	Wired	Tests	Valida
docker-compose.yml — 2-container default + Go profile + boba-jackett service	infra	yes	e2e test_live_containers.py	tested-green-suite
Jackett auto-configuration at startup (key extract → inject JACKETT_API_KEY)	implemented	yes	integration test_jackett_autoconfig_real.py	tested-green-suite
BOBA_MASTER_KEY auto-generation (first-boot) for encrypted boba.db	implemented	yes	go bootstrap/bootstrap_test.go	unit (g

Components NOT inventoried / discrepancies (honest gaps, §11.4.6)

- **CORRECTION vs Rev 2:** boba-ctl.sh DOES exist at scripts/boba-ctl.sh (Rev 2 said it did not). Its subcommands are now itemized in §8a. The underlying Go binary is cmd/boba-ctl/main.go (5 subcommands: up/down/status/health/list).
- **CORRECTION vs Rev 2:** the install-plugin.sh PLUGINS curated array has **44 entries**, not 12. ~21 have a matching plugins/*.py file; the rest (incl. torrentproject, torrentscsv) are curated names with no file present in this tree — itemized in §4c, not asserted as working.
- **Prometheus/OpenTelemetry metrics** — tests/observability/test_metrics_exist.py + observability/ dir exist, but no prometheus_client Counter()/Histogram()/Gauge() definitions were found in download-proxy/src; metrics surface is via GET /api/v1/stats (counters maintained in-process), not a /metrics Prometheus endpoint. Stated as fact, not assumed.
- **Go DownloadHandler / ActiveDownloadsHandler** — flagged partial/stub in §2a (mock success / empty array, no qBit integration) per source reading. The Go profile is opt-in and NOT the shipped product.
- **Go scheduler** — CRUD-only, no driver loop (RW-10); schedules never fire. The Go enricher is missing (RW-11).

- **Video-recording confirmation** — PENDING for all rows EXCEPT the boba-ctl CLI subcommands (status/health/list) and the web-UI flows already filmed (POST /api/v1/search journey + dashboard tab/button/theme/Jackett tour). Remaining per-feature flows are an in-progress recording pass (§11.4.107/§11.4.143).